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Project Title:	OASIS: Optimizing Auditory Stimulation to Improve Cognitive Performance Using SmartSleep		
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Task Description:	Simply put, individuals who do not get enough sleep do not perform well when awake. Although physical rest is an important part of the benefit gained from sleep, adequate sleep is essential for optimal brain function. Not surprisingly, inadequate sleep is a great concern for astronauts during spaceflight missions. Astronauts often get less nightly sleep than they should (< 7 hours) and may be forced to get little sleep for prolonged periods. Unfortunately, taking sleeping pills is often counterproductive. In addition to other unwanted side effects, like addiction and increased tolerance with prolonged use, sleep medications often lead to daytime grogginess and a lack of mental alertness, the opposite of the mentally refreshed feeling sleep is supposed to provide. Our group has spent over a decade developing a technique to optimize sleep without using medications. We designed a system that capitalizes on two of the most important things the brain needs to do during sleep - produce slow waves and monitor the environment. Slow waves are the dominant rhythm of the brain when it is in its deepest and most restorative sleep state. When the brain goes offline during sleep, slow waves work to actively restore the brain in preparation for the next day. However, the act of going offline during sleep is dangerous for the brain because it can no longer do one of its primary functions - protect the body. Therefore, even during deep sleep, the brain subtly monitors the surrounding environment, most effectively by listening for sounds. The device we developed takes advantage of this fact and actively monitors the brain during sleep to reliably detect slow waves. When slow waves are detected, auditory stimuli are played at a frequency that results in a robust increase in slow waves, the most important restorative rhythm of sleep. However, the overall benefits of enhancing sleep slow waves with auditory stimulation has not been adequately tested. Largely, this is because an adequate system for nightly stimulation has not been available. While several research groups have recently demonstrated improved memory retention in the laboratory after playing appropriately timed sounds during slow wave sleep, these effects were demonstrated using a different stimulation protocol than the one we developed and typically did not show an overall increase in sleep slow waves across the night. This suggested that other than improved memory, the stimulation might not affect the other known restorative benefits of sleep. Assessing the cognitive effects of tone-enhanced sleep is also challenging in the laboratory due to the variability in sleep, between different people, and even night-to-night within the same person. On any given night in the laboratory, sleep might be better and subsequent cognitive performance might improve but this will have nothing to do with the stimulation and will have more to do with chance. Maximizing the sleep enhancing benefit of a system means adapting to and accounting for an individual and his/her changing sleep patterns and testing the subject for longer periods of time, not just a few nights. The ability to enhance sleep with a wearable device that is comfortable, easy to use, and continuously adapts to an individual is a truly exciting prospect. The proposed study will be the first to assess the ability of different ways of playing auditory tones during sleep to enhance slow waves and improve cognitive performance over a long-term period (8 weeks). We will record and stimulate subjects in their home environment using the recently developed, commercially available wearable device and test subject performance throughout each day using a short-term test battery developed particularly for astronauts. This proposal represents the vital first steps towards optimizing auditory stimulation during sleep to enhance long-term cognitive performance using a device that is easily translatable to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Individuals who do not get enough sleep do not perform well when awake whether in space or on Earth. Although physical rest is an important part of the benefit gained from sleep, adequate sleep is essential for optimal brain function. Unfortunately, taking sleeping pills is often counterproductive. In addition to other unwanted side effects, like addiction and increased tolerance with prolonged use, sleep medications often lead to daytime grogginess and a lack of mental alertness; the opposite of the mentally refreshed feeling sleep is supposed to provide. The ability to enhance sleep with a wearable device that is comfortable, easy to use, and continuously adapts to an individual is a truly exciting prospect. Taking advantage of SmartSleep headband technology, an innovative, non-invasive, closed-loop brain stimulation and recording technology, this is the first ever study to directly evaluate the ability of various auditory stimulation protocols to enhance sleep and cognitive function in an ecological, “at-home” sleep environment over the course of several weeks. This study has the potential to transform sleep health during spaceflight and on Earth by (1) determining how distinct methods of auditory slow-wave enhancement impact neurobehavioral performance daily and (2) advancing individualized sleep enhancement to meet the next day task demands. The insights gained from this immense data set will help us to further improve and individualize sleep enhancement and provide further insights into the link between sleep quality and cognitive performance in a way previously unachievable.
	During and even before spaceflight missions, astronauts may not have adequate opportunity for sleep, yet they are required to maintain alertness for extended periods. Even modest sleep restriction takes a heavy toll on reaction time, motivation, attention, memory, endurance, and judgment. Therefore, when opportunities for sleep are available, it is essential that sleep is as efficient as possible. Accumulating evidence suggests many of the benefits of sleep are associated with sleep slow waves, the near-synchronous alternation between depolarization and hyperpolarization by millions of cortical neurons occurring approximately once per second during non-rapid eye movement (NREM) sleep. Slow waves are thought to mediate the restorative effects of sleep on brain networks and cells by promoting specific patterns of neuromodulatory and electric activities. Pharmacologic approaches to improve sleep efficiency by increasing slow waves have been relatively unsuccessful. Moreover, such agents remain in an astronaut’s system if they are awoken from sleep, needing to function, and are often associated with residual waking side effects. Recent data from several different laboratories has now confirmed that low intensity (sub-arousal) auditory tone stimulation, when given regularly at approximately 1 Hz during NREM sleep, can consistently enhance slow waves and some aspects of cognitive performance. This type of stimulation is not only effective, but also safe, controllable, and able to be non-obtrusively administered during sleep. Several groups have shown positive next-day effects on memory retention after auditory stimulation enhanced sleep. Despite these common successes, the auditory stimulation protocols used by different laboratories vary substantially. Moreover, assessment of the long-term impact of auditory sleep enhancement paradigms on sleep and performance has never been thoroughly examined because it would require a user-friendly, flexible platform for stimulation, comfortable enough to be worn routinely. In this study, we used a sleep-wearable device (SmartSleep), developed by our group, in collaboration with Philips Healthcare, that monitors sleep electroencephalography (EEG) in real time and can easily be managed by a user in their home sleep environment. To enhance sleep, the device delivers auditory stimulation in the form of 50-millisecond tones during non-REM sleep in a closed-loop fashion at a volume dynamically modulated by sleep depth. Three different auditory stimulation protocols were tested for two-week periods with the order randomly assigned and blinded to the

Task Progress:	subjects. Each day, subjects took the Psychomotor Vigilance Test (PVT) and a series of validated behavioral questions that assessed sleepiness, physical fatigue, and mental tiredness 30-60 minutes after awakening, filled out a Karolinska Sleepiness Scale three times (in the morning, afternoon, and evening), and performed a full Cognition test battery approximately one hour before sleep. A no-stimulation (SHAM) week occurred between each of the 2-week stimulation periods. During the sham weeks, participants continued to perform the daily neurocognitive testing. The aims of the study were to (1) confirm in a group of astronaut-like subjects that sleep and cognitive function could be enhanced in an ecological setting using closed-loop auditory stimulation during sleep, assessed with objective and subjective sleep measures and a validated cognitive test battery; (2) determine the extent to which various closed-loop auditory stimulation protocols differentially affected various aspects of sleep and cognition in individuals; (3) explore whether an individual's previous night's sleep EEG metrics could predict cognitive performance; and (4) assess any potential adverse effects of long-term slow wave sleep enhancement. Due to the extraordinary circumstances surrounding data collection, we also added an unplanned exploratory aim examining the impact of COVID on sleep and cognition. Data collection began September 30, 2019. Prior to being halted due to COVID-19 pandemic restrictions in March 2020, 13 subjects completed the study. After data collection resumed, in June 2020, 15 more subjects completed prior to the study end in December 2020. The final sample of N = 28 subjects was 4 more than were originally planned and also included a similar number of subjects before (N = 13) and during (N = 15) the COVID-19 crisis, allowing for the exploratory group level analysis of the pandemic's impact on sleep and cognition as referenced above. Over the course of the 2 year study, we collected 1147 overnight, single channel, sleep EEG recordings, 1123 morning PVTs (98%), 3266 Karolinska Sleepiness Scales (KSS), 95%, and 1137 evening cognition batteries (99%). On 70 nights (6%) there were device technical issues that resulted in significant signal loss or poor quality signal. Thirty-five nights (3%) were interrupted, had subject compliance issues or needed to be redone because of subject scheduling issues. On only 8 nights (<1%) did subjects report hearing tones. Thirty-three nights (3%) were recorded as makeup nights on a weekend (Friday or Saturday). Out of 1120 expecting sleep EEG recording nights expected (40 nights/subject; 2 weeks each x 4 conditions), we were able to successfully record 1014 nights (91% success) excluding makeups. In association with the completed nights, subjects completed 989 Morning PVTs, 2932 KSSs, and 998 Evening Cognition batteries (including PVT), all collected in the field (i.e., at home). No adverse events were reported. Preliminary examination and communication with subjects during the study revealed that subjects reported sleeping comfortably with the device and only rarely reported any discomfort at all, seemingly unassociated with stimulation. Intent to treat analysis, along with in-depth data quality and integrity assessments are ongoing.
Bibliography Type:	Description: (Last Updated: 08/04/2022)
Abstracts for Journals and Proceedings	Riedner BA, Garcia-Molina G, Peterson BF, Jones SG, Myers AM, Jones CW, Basner M, Dinges DF, Tononi G. "OASIS: Optimizing auditory stimulation to improve cognitive performance using SmartSleep." 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021